

202044502 Programming with JAVA

Unit -1 Topic – O.O, Class, Object

Academic Year: 2024-25 (II)

Prepared By:
Prof. Pratik B. Soni
(Asst. Prof. – I.T. Dept, MBIT)

Control Statements

- **if**
- **nested -if**
- **else – if ladder**

```
int a, b;  
// ...  
if(a < b)  
    a = 0;  
else  
    b = 0;
```

```
if(i == 10)  
{  
    if(j < 20)  
        a = b;  
    if(k > 100)  
        c = d;  
  
    else  
        a = c;  
}  
Else  
    a = d;
```

```
class IfElse  
{  
    public static void main(String args[])  
    {  
        int month = 4; // April  
        String season;  
  
        if(month == 12 || month == 1 || month == 2)  
            season = "Winter";  
  
        else if(month == 3 || month == 4 || month == 5)  
            season = "Spring";  
  
        else if(month == 6 || month == 7 || month == 8)  
            season = "Summer";  
  
        else if(month == 9 || month == 10 || month == 11)  
            season = "Autumn";  
  
        else  
            season = "Bogus Month";  
        System.out.println("April is in the " + season + ".");  
    }  
}
```

Control Statements

- **switch**

```
class SampleSwitch
{
    public static void main(String args[])
    {
        for(int i=0; i<6; i++) {
            switch(i) {
                case 0:
                    System.out.println("i is zero.");
                    break;
                case 1:
                    System.out.println("i is one.");
                    break;
                case 2:
                    System.out.println("i is two.");
                    break;
                default:
                    System.out.println("i is greater than 3.");
            }
        }
    }
}
```

Control Statements

- **while**
- **do - while**

```
public static void main(String args[]) {
    int n = 10;

    while(n > 0) {
        System.out.println("tick " + n);
        n--;
    }
}
```

```
class DoWhile {
    public static void main(String args[]) {
        int n = 10;

        do {
            System.out.println("tick " + n);
            n--;
        } while(n > 0);
    }
}
```

Control Statements

- for

```
class ForTick {  
    public static void main(String args[]) {  
        int n;  
  
        for(n=10; n>0; n--)  
            System.out.println("tick " + n);  
    }  
}
```

Control Statements

- for
- nesting loops

```
class Nested {  
    public static void main(String args[]) {  
        int i, j;  
  
        for(i=0; i<10; i++) {  
            for(j=i; j<10; j++)  
                System.out.print(".");  
            System.out.println();  
        }  
    }  
}
```

Control Statements

- **Break** – as it executes loop will be terminates, and go for next statement.
- **Continue** – as it executes loop will go for next iteration.

```
class BreakLoop {
    public static void main(String args[]) {
        for(int i=0; i<100; i++) {
            if(i == 10) break; // terminate loop if i is 10
            System.out.println("i: " + i);
        }
        System.out.println("Loop complete.");
    }
}
```

```
i: 0
i: 1
i: 2
i: 3
i: 4
i: 5
i: 6
i: 7
i: 8
i: 9
Loop complete.
```

Control Statements

- **Break** – as it executes loop will be terminates, and go for next statement.
- **Continue** – as it executes loop will go for next iteration.

```
// Demonstrate continue.
class Continue {
    public static void main(String args[]) {
        for(int i=0; i<10; i++) {
            System.out.print(i + " ");
            if (i%2 == 0) continue;
            System.out.println("");
        }
    }
}
```

```
0 1
2 3
4 5
6 7
8 9
```

Control Statements

- **Return – return with / without value back to caller.**

```
// Demonstrate return.
class Return
{
    public static void main(String args[])
    {
        boolean t = true;

        System.out.println("Before the return.");

        if(t)
            return; // return to caller

        System.out.println("This won't execute.");
    }
}
```